5 – Writing Rubrics

MLI Rubrics Project – Lesson 5.mp4

Objective

So in this lesson, we're going to walk you through:

- What a rubric is and why it matters
- The parts that every rubric must include
- How to write each part effectively
- And examples of strong rubrics.

By the end, you'll know exactly how to build a rubric that makes it clear what counts as a successful, expert-level answer. In the lesson after this, we will go over common mistakes to avoid, examples of unsuccessful rubrics, and go over other common issues that often arise.

You can find examples of rubrics by domain in the project guidelines document.

What is a Rubric?

When you hear the word "rubric," you probably think of how teachers grade assignments – a list of expectations or checkboxes that determine whether a student did the assignment well. That's the basic idea here too – but rubrics for training AI models do have a very specific format that will feel new and different.

A Rubric is a Scoring Guide for AI

For training AI models, rubrics are **a list of precise, yes-or-no, true-false criteria** that tells us whether an AI model's response to a question is good or bad – and to what extent. The rubric is like a **step-by-step checklist** that lets a human reviewer score the AI's response objectively and consistently, without needing to guess or make subjective judgments.

So you can think of it like grading with an answer key – but for the complex tasks that you have to do in your job. Let's say you're asking an AI model to calculate how many samples of two products you can buy with a leftover budget after sponsoring a music festival. The rubric would spell out, in clear steps:

- Did the model correctly identify the largest festival under budget? **Yes/No**
- Did it correctly calculate how much budget is left? **Yes/No**
- Did it compute how many samples could be purchased? **Yes/No**
- Did it mention relevant laws that make the entire task invalid? **Yes/No**

Without a rubric, we have no clear or consistent way to evaluate whether the model did the task correctly – especially when the task is complex or requires multiple steps.

So a well-designed rubric:

- **Makes scoring consistent**, even across different human reviewers.
- **Clarifies what matters most** in a good answer.
- **Teaches the AI** what success looks like when it's being trained.

So your job will be to write a rubric that:

- Fully captures what a perfect answer requires.
- Breaks that answer into smaller, clear components that can each be scored true/false.
- Uses sources and evidence to justify each criterion.

If you do this well, any competent reviewer could read your rubric, look at the AI's response, and confidently score it – without needing to do any extra research.

How to Generate the Criteria for Your Rubric

So now let's look at exactly how to build the **rubric criteria.** The criteria are the backbone of your rubric: they define, step by step, everything that needs to appear in a perfect response.

But how do you actually come up with that list? Here's a **simple, step-by-step process** to help you generate clear, complete criteria for any prompt.

Step 1: Start with your Golden response

While we previously used scaffoldings to jumpstart the rubric writing process, you can now use the golden response to autogenerate a set of criteria. However, it's important that you think about what criteria would be needed to judge any high quality response to this topic, not just the one you provided. This is the most important first step – that before you start writing criteria, you imagine that you're completing the task yourself by actually writing out your own full answer and **mapping out the steps you would take to arrive at a perfect answer.** Your criteria will ultimately need to have all of the components that an ideal answer to the prompt that you have made would require.

Step 2: Break the Answer Into Discrete Components to Create Your Criterion Description

Next, break down your complete answer into **small, individually checkable criteria.** This step is where we get granular and incredibly specific. This is what we call the **Criterion Description** on your submission – and it will be the one-sentence description of the rubric criteria. **You will**

need at least 10 distinct criteria and you can have no more than 30. Your criteria descriptions must meet the following guidelines:

- They should be **falsifiable:** Each component should be something that can be graded as **true or false** – i.e., a reviewer using your rubric and looking at an AI model's response to the prompt should be able to easily say, **"Yes, the model did this,"** or **"No, it didn't"** without having to search for more information or additional context to make the determination.
    - So a criteria like "Identifies common side effects of Medication Y" would be too vague because it's not spelled out what those side effects are.
    - Instead, you should say something like "Identifies at least two common side effects of Medication Y: nausea, headache, dizziness, fatigue, or skin rash."

- **Self-contained:** The criteria description also needs to be fully **self-contained**. When we say that criteria must be self-contained, we mean that each individual criterion should include everything the reviewer needs to evaluate whether the AI got that point correct – without doing any extra research.
    - Imagine if your prompt is "Provide a list of all restaurants within 1 mile of 123 Main St, San Francisco, CA" and a rubric criteria said "mentions all restaurants within 1 mile of 123 Main St" – the issue here is that criteria does not explicitly list all the restaurants within 1 mile of 123 Main St. If you're a reviewer using this rubric and looking at the model's response to the prompt question, you'd have no way of evaluating whether or not the model got it right because "mentions all restaurants" is just too vague.
    - The bottom line is that if the reviewer can't tell whether the AI is right just by looking at your criterion (and its source and quote), then the criterion is **not self-contained.**
    - So any criteria like "correctly calculates the answer" aren't going to cut it – they're too vague for helping us to evaluate whether the AI models are answering the prompts correctly. This is another way of saying that your criterion description must be **clear and unambiguous.**

- They should convey **universal expert agreement** – the grading criteria should be universally accepted by peers in your field. Any component professional should be expected to pass all your criteria.
    - That means you should avoid tasks or criteria with highly subjective interpretations, such as "why did a major historical event occur" where experts might fundamentally disagree.
    - This also means that you should avoid criteria that favor one valid approach over other equally valid alternatives.

Step 3: Prioritize – What Matters Most?

Not all parts of the answer are equally important. You should ask yourself: which criteria are most important? How would I categorize their importance?

You will have to **assign each criteria a weight** representing its relative importance. The weighting categories are below. These have changed recently, and so may differ from the videos. If criterion are not important enough to meet one of the qualifications below, they should be excluded:

1. **Primary objective:** a direct response to the questions asked in the prompt. Without this piece of information, the response is incorrect or incomplete
2. **Not primary objective:** a piece of information that is not explicitly asked for in the prompt, but that materially improves the quality of the response

Step 4: Support Each Criterion with a Source and a Rationale

For each criterion, you need:

1. A **source link to the supporting document** where the necessary information for this criteria can be found. You already learned about collecting all your sources and materials in the last lesson! Here, you're just showing exactly how you'd be using that evidence yourself if you were answering the prompt. The source for a parent criterion **should not** be included in a child criterion unless that source must again be referenced for that step.

2. A **rationale** that pulls a specific quote or data point from your source that directly supports the criterion. The quote should clearly contain the information the AI is expected to include. If there is no quote but there is analysis required, simply state the formula or calculation you would use there. Think of the rationale as "showing your work" in a way that makes it easy for reviewers to check whether the AI's response to your prompt matches the fact in the source that supports your criteria.

Step 5: Categorize the Criterion Type

You will also need to categorize what type of criteria you have created. Here are the three types:

- **Extraction:** Identifying facts and claims from the supporting documents that are needed to answer the prompt. In some cases, these facts are only required as inputs for model reasoning.

- **Reasoning:** Performing inductive, mathematical, and common sense reasoning to reach new analyses and conclusions based on the source documents. Here are some examples:
    - "Calculates the percentage of paid taxes in the bill as 30%" – 30% is not retrieved from a source, which means the model has to perform math to calculate it based on previously determined values (e.g. meal price and total bill price).
    - "Ranks cars by MPG as car 1> car 2> car 3" – the model uses the MPG values to build a rank.

    - “Mentions the blood test will show elevated glucose levels” – this model uses previous information to reach this conclusion (e.g., “the patient is diabetic,” “the patient had no insulin access,” or “the patient just had a high sugar meal”).

- **Style:** Matching requirements for style, format, and tone. For instance, this could include following stylistic rules on how to deliver financial advice.
    - “The output includes a table summarizing FY2024 values” – the model follows the request to build a table in the prompt.

All the criteria you write must have at least one criterion type for categorization purposes – but if a criteria falls under multiple types, you may include all relevant types for that criteria.

Step 6: Consider Criteria Dependency

Sometimes, a criterion in your rubric relies on information from another step. When this happens, the second criterion is considered **dependent** on the first. Let’s break down what that means and how to identify these relationships.

A **dependent criterion** is one that can’t be answered or evaluated on its own – it requires the answer or outcome of another criterion first. For example, if one criterion asks the model to identify the most sold artist in the U.S. in 2024, and another asks for that artist’s age, the second one depends on the first. You can't evaluate the second unless the first has been done correctly.

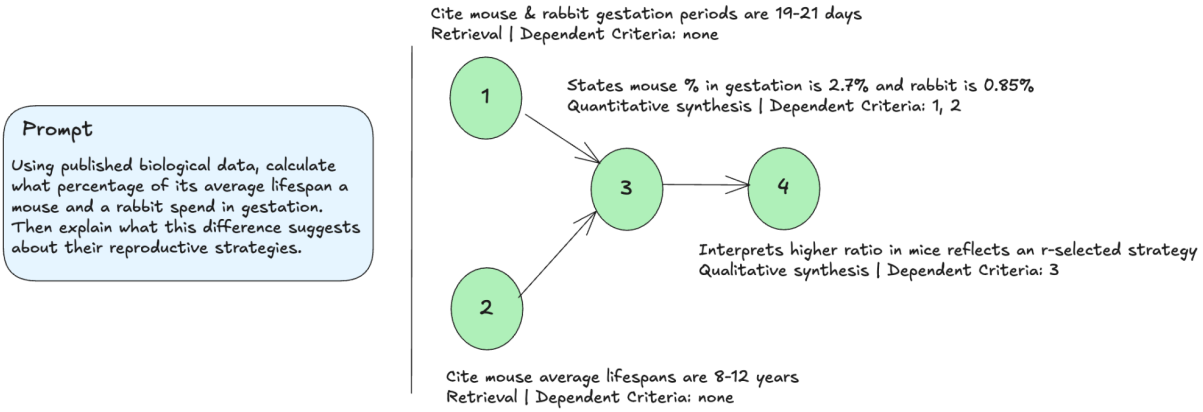
Key Guidelines:

- **Each dependent criterion still needs a type.** It could involve extraction, reasoning, or style – dependency is a relationship, not a replacement for type.
- **A criterion can have more than one dependency.** If it’s built on multiple prior steps, list them all.
- **Only mark direct dependencies.** Don’t trace back the full chain. If Criterion 4 depends on 3, and 3 depends on 1 and 2, then Criterion 4 should only list 3 as a dependency—not 1 and 2.

When Is a Criterion *Not* Dependent?

- If a piece of information can be pulled directly from a source without needing earlier steps, it’s *not* a dependent criterion.
- **Exception:** If the model has to *choose* something (e.g., pick a target company) and then gather facts *about* that choice (e.g., stock price, headquarters), those fact-based criteria are still dependent on the initial choice.

Here’s an example:



Conclusion

When your draft list is done, you should ask yourself: If a model included everything in this rubric, would that be a complete, correct answer? If so, that's a sign that you probably made a strong rubric!

We're not quite done with rubrics yet though – in the next lesson, let's switch gears and go over common mistakes that people make and how to avoid them.